

NeuroShift vs Base Paper

Easy comparison draft for guide discussion | July 2026

The base paper improves how reliably we recognize a gesture. NeuroShift uses gaze to select an appliance and EMG to confirm the toggle, or abstains when unsure.

1. One-line summary

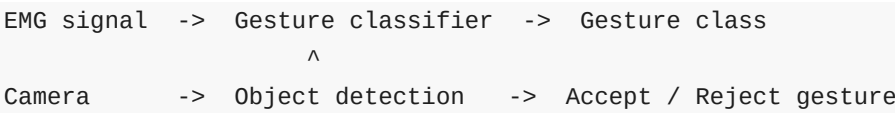
	Base paper	NeuroShift (our work)
Idea	Use camera to make EMG gestures more reliable	Gaze selects appliance; EMG confirms toggle; Act/Abstain

Base paper: Better gesture recognition.
Our paper: Safer smart-home intention understanding.

2. Base paper

Title: Visual Scene Understanding for Enhanced EMG Gesture Recognition
Authors: Chamberland, Labbe, Tam, Scheme, Gosselin
Venue: IEEE EMBC 2024
DOI: 10.1109/EMBC53108.2024.10782354

What they do



- User wears EMG sensors.
- System classifies hand gestures.
- Camera detects objects in the scene.
- Vision checks if the gesture makes sense (reduce false movements).
- Target: myoelectric prosthesis control.

What they solve

- False / unstable gesture predictions during onset and holding.
- More robust EMG control using scene context.

What they do NOT solve

- Smart-home appliance control
- User intention beyond predefined gestures
- Act vs Abstain for safety in multi-object rooms
- Personal habit / routine learning
- Caregiver explanation of why a device fired

3. Our idea - NeuroShift

Working title: NeuroShift: Affordance-Conditioned Multimodal Intention Inference with Selective Actuation for Assistive Smart Homes

What we do

```

Camera -> YOLO appliances      -> Devices in scene
Camera -> Gaze / head direction -> Which device is selected
EMG     -> Volition score       -> Confirm real intent?
Context -> Time / habit (Sem 8) -> Reweight if needed

      |
      | Intention Policy
      |
      | Act / Abstain / Clarify
      |
      | ESP32 + MQTT -> Lamp / Fan / Plug
    
```

- Camera finds appliances AND which one the user is looking at.
- EMG confirms the user truly wants to toggle it.
- Looking alone never actuates (Midas-touch protection).
- Target: assistive smart home for motor-impaired users.

4. Side-by-side comparison

Point	Base paper	NeuroShift
Research question	How to make EMG gestures more robust with vision?	How to infer home-device intention and avoid wrong actuations?
Domain	Prosthesis / gesture control	Assistive smart home / independent living
Input	EMG + camera	EMG + camera + context (Sem 8)
What AI predicts	Gesture class	Device action or abstain
Role of vision	Validate / rescue gesture	Detect appliances + gaze-select target
Role of EMG	Classify gesture	Confirm toggle intent only
Selector	Hand near object	Head/eye gaze (primary)
If scene is confusing	Gesture correctness focus	System can ABSTAIN (safer)
Main metric	Gesture accuracy / robustness	False Actuation Rate + intention F1
User burden	Learn a gesture set	Look + confirm with muscle effort
Hardware	Lab prosthesis setup	MyoWare + webcam + ESP32 relay hub

5. Simple analogy

Analogy	Base paper	NeuroShift
Like...	Spell-checker for hand gestures	Smart assistant for what you want to do at home

NeuroShift 2.0 | Base Paper Comparison

Example	User did a fist - is that valid near this object?	Lamp + fan visible; EMG shows effort - turn lamp or wait?
Failure mode	Wrong gesture / false movement	Wrong device turns on / unsafe actuation

6. Our novelty

We are NOT claiming

- First paper to combine EMG + camera
- Highest gesture accuracy on Ninapro
- A new Transformer just for EMG

We ARE claiming

- From gestures to home intentions: appliance action, not gesture class.
- Selective actuation (Act / Abstain): prefer no action over wrong action.
- Sparse EMG as confirmation, not a gesture dictionary.
- Personal context + explanations in Sem 8 for caregivers.

7. Extension statement

Chamberland et al. (EMBC 2024) demonstrated that visual scene understanding can enhance EMG-based gesture recognition by reducing false movements in prosthesis control. NeuroShift extends this multimodal paradigm from gesture validation to assistive smart-home intention inference. Instead of recognizing predefined gestures, our system combines ambient visual affordances with sparse EMG volition and personal temporal priors to decide which device to actuate or whether to abstain, targeting reduced false actuations for motor-impaired independent living.

8. Two-semester plan

Semester	Phase	Deliverables
Sem 7	Phase 1 - Prototype	EMG + vision + MQTT + logging + 3 baselines
Sem 8	Phase 2 - Research	Fusion, Act/Abstain, prior, XAI, evaluation, IEEE paper

9. Flowchart comparison

Base paper

```
EMG --> Gesture model --> Gesture
      ^
Camera --> Objects -----> Gate (accept/reject)
```

NeuroShift

```

Camera --> Objects + Pose --> Candidate devices
EMG      --> Volition score --> Confirm effort
Context--> Habit prior      --> Reweight options (Sem 8)
      |
      Intention Policy
      |
      Act / Abstain / Explain
      |
      Smart home device

```

10. Closing line

The base paper improves how reliably we recognize a gesture. NeuroShift uses gaze to select an appliance and EMG to confirm the toggle, or abstains when unsure. That is our novelty.